

HONG KONG BAPTIST UNIVERSITY

Audio and Speech Processing

Practical Assignment II

Introduction

This practical assignment is designed to assess students' understanding and programming skills in Machine Learning. Students will be tested on their replication and implementation of *dimensional reduction* and *classification analysis* of machine learning methods. Bonus scores will be provided for students for their innovative programming on an extra task with additional feature construction. They are also required to report results by showing critical thinking in interpreting the results.

Program Code Template

The basic program codes by Python are provided for your reference in the document “*audio_2.ipynb*”. All the tasks can be implemented based on the given codes with slight adjustments of commenting certain functions, changing parameters, or features. Google Colab and PyCharm can be the platform for you to execute the programming.

Tasks

- **Basic Task:** Replicate the given codes in the template file and successfully run the codes on your given subset of data. Report the implementation results in your *.docx* file. Provide basic descriptions of the results.
- **Task 1:** Generate the waveform, spectrogram, and MFCC figures on the given data (according to your student ID). For spectrogram, set the window length be 256, and hop length be half of the window length. Comment on the difference between the figures and sound effects. Report your findings.
- **Task 2:** Feature reduction. Implement the PCA on features. Generate the feature visualization image. Report on how these changes affect visualization.
- **Task 3:** Classification effect. Compare the classification accuracy after feature reduction has been applied (PCA components = 10) before KNN and SVM methods. Report your findings.
- ***Extra/bonus Task:** Re-express each task code above as function format. Specify the inputs and outputs of the functions. Report your results in report and “*code_improved.ipynb*”.

Submission

1. Submit ONE *.ipynb* file containing all your program codes for tasks. Rename the *.ipynb* file to your name, e.g. ChanTaiMan.*.ipynb*. Submit the bonus problem files separately.
2. Submit ONE word file reporting your results and findings. Rename the *.docx* file to your name, e.g. ChanTaiMan.*.docx*.

Marking Criteria

Items	Evaluation Standard	Marks
Basic Task	Replication of running codes on given data	10
Task 1	Success in running task 1 and report the result	20
Task 2	Success in running task 2 and report the result	20

Task 3	Success in running task 3 and report the result	20
Report	Organized, elaborative, well-formatted	15
Analysis	In-depth thinking and reflections	15
Total:		100
Bonus:	Innovative programing and analytical skills	20

Appendix A

You should work on one of the corresponding subset data as listed in Table 1.

Last digit of student ID	Name of subset data
0-4	Blue, Country, and Pop
5-9	Classical, Jazz, and Rock

Table 1 - Mapping of Student ID and Subset Number.